

AMIT KUMAR
POSTDOCTORAL RESEARCHER,
BIOINSPIRED SOFT ROBOTICS LAB,
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OBJECTIVE

Postdoctoral researcher developing adaptive biomaterials and environmentally responsive soft robotic systems through bioinspired material design, 4D printing, and stimuli-responsive polymers for biomedical and agricultural applications.

RESEARCH AREA

My research area includes:

- **Functional Responsive Materials:** Hydrogel-, elastomer-, and nanoparticle-based systems with solvent and light programmable actuation.
- **Interfacial Polymer Chemistry:** Engineering dynamic biomaterials with biocompatible crosslinking strategies, bilayer architectures, and particle self-assembly to develop a novel multimaterial system for medical applications.
- **3D/4D Printing of Soft Materials:** Direct ink writing, bath-assisted printing, and molding of architectures that transform over time under environmental or external stimuli.
- **Soft Robotics:** Designing biodegradable, adaptive soft robotic systems for autonomous actuation and controlled molecular delivery in biomedical and agricultural applications.

EDUCATION

- **DOCTOR OF PHILOSOPHY (PH.D.) + MASTER OF SCIENCE (M.S.)- 9.75 (CGPA)** JULY'17 - JULY'24
Dual Degree Interdisciplinary Research Program,
Department of Biotechnology, Indian Institute of Technology Madras, Chennai, India
Thesis: Harnessing Solvent-Responsive Actuation Behavior of Polymers for Biomedical Applications
Supervisors: Prof. Nitish R. Mahapatra, Department of Biotechnology,
Prof. Pijush Ghosh, Department of Applied Mechanics & Biomedical Engineering
- **BACHELOR OF TECHNOLOGY (B.TECH.)- 70.42 (CPI)** AUG'13 - MAY'17
Department of Biotechnology, Delhi Technological University, Delhi, India

PROFESSIONAL EXPERIENCE

- **POSTDOCTORAL RESEARCHER, ISTITUTO ITALIANO DI TECNOLOGIA (IIT), ITALY** 1ST JUNE'25 - PRESENT
Mentor: Prof. Barbara Mazzolai, Director, Bioinspired Soft Robotics (BSR) Laboratory, IIT, Genoa
- **RESEARCH ASSOCIATE, INDIAN INSTITUTE OF TECHNOLOGY MADRAS (IITM), INDIA** SEPT'24 - FEB'25
Mentor: Prof. Pijush Ghosh, Nanomechanics Lab, Department of Applied Mechanics & Biomedical Engineering, IITM, Chennai
- **INDUSTRIAL TRAINEE, NATIONAL PHYSICAL LABORATORY (NPL), CSIR, DELHI, INDIA** JUNE'16 - AUG'16
Mentor: Dr. G. Sumana Gajjala, Biomolecular Electronics and Conducting Polymer Research Group, NPL, CSIR
- **INDUSTRIAL TRAINEE, INMAS, DRDO, DELHI, INDIA** DEC'15 - JAN'16
Mentor: Dr. Aseem Bhatnagar, Department of Nuclear Medicine, INMAS, DRDO

TECHNICAL SKILLS

- **LABORATORY PROFICIENCIES:**
 - **Responsive biomaterials fabrication:** Hydrogels, Elastomers, Nanoparticles, Bilayers, 3D printing (DIW)
 - **Advanced microscopy and spectroscopy:** SEM, FT-IR, UV-Vis
 - **Cellular assays:** MTT, live/dead bacterial viability, antibacterial studies, biodegradability studies
- **COMPUTATIONAL AND ANALYTICAL TOOLS:**
 - **Visualization & Design tools:** Origin, Graphpad Prism, Chemdraw, Inkscape, Blender, Filmora 9.0, Ultimaker Cura
 - **Documentation:** Microsoft Office, LaTeX

RESEARCH PUBLICATIONS

1. Anas Saifi, Smruti Parimita, **Amit Kumar**, Pijush Ghosh. Programmable 4D-Printed Soft Actuators: Harnessing Bending Strain Distribution for Embedded Topological Functionality. *ACS Applied Engineering Materials*, (2025). (IF- 3.5)
2. **Amit Kumar**¹, Shaik Sameer Basha¹, Mukilarasi B, Vimalraj Selvaraj, Pijush Ghosh, Swathi Sudhakar. Leveraging Cancer Microenvironment pH Shift for Actuation-induced Multidrug Release from Chitosan/Carboxymethyl Cellulose Bilayer Film. *International Journal of Biological Macromolecules*, (2025). (IF- 8.7)
3. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy, Pijush Ghosh. 4D printing of pH-responsive bilayer with programmable shape-shifting behaviour. *European Polymer Journal*, (2024). (IF- 6.8)
4. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra, Pijush Ghosh. Superhydrophobic Asymmetric pH-responsive Soft Actuators: Implications for the Development of Anti-fouling Medical Devices. *Chemical Engineering Journal*, (2024). (IF- 12.5)
5. Nidhi Gupta¹, **Amit Kumar**¹, Pravin K Vaddavalli, Nitish R. Mahapatra, Akhil Varshney, Pijush Ghosh. Efficient Reduction of the Scrolling of Descemet Membrane Endothelial Keratoplasty Grafts by Engineering the Medium. *Acta Biomaterialia*, (2023). (IF- 10.4)
6. Smruti Parimita, **Amit Kumar**, Hariharan Krishnaswamy, Pijush Ghosh. Solvent triggered shape morphism of 4D printed hydrogels. *Journal of Manufacturing Processes*, (2023). (IF- 7.8)
7. **Amit Kumar**, Raja Rajamanickam, Joyita Hazra, Nitish R. Mahapatra, Pijush Ghosh. Engineering the Nonmorphing Point of Actuation for Controlled Drug Release by Hydrogel Bilayer across the pH Spectrum. *ACS Applied Materials & Interfaces*, (2022). (IF- 7.8)

RESEARCH PUBLICATIONS (TO BE COMMUNICATED/UNDER REVIEW)

1. **Amit Kumar**, Anas Saifi, Kumari Kiran, Sanjib Senapati, Nitish R. Mahapatra, Pijush Ghosh. Dual-Responsive Bidirectional Actuator for Biofilm Disruption on Medical Surfaces. *Advanced Healthcare Materials*- (Under Review)
2. Ruowen Tu¹, **Amit Kumar**¹, Barbara Mazzolai. An Application-Driven Framework for Designing and Assessing Biodegradable Soft Robots. *Perspective*- (To be Submitted)
3. Yesheneh Jejaw Mamo, Ruowen Tu, **Amit Kumar**, Yasmin Tauqeer Ansari, Barbara Mazzolai, Nicola M. Pugno. Hyperelastic Hierarchical Origami Actuators Powered by Hydrogel Swelling. *Article*- (To be Submitted)

POSTERS AND PRESENTATIONS

1. **Amit Kumar**, Smruti Parimita, Kumari Kiran, Nitish R. Mahapatra and Pijush Ghosh, "Superhydrophobic Asymmetric Solvent Responsive Actuator for Antifouling Medical Devices", *Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza*, 5th July 2024, IIT Madras, Chennai, India.
2. **Amit Kumar**, Nidhi Gupta, Nitish R. Mahapatra and Pijush Ghosh, "Strategies to attain underwater actuation of solvent-responsive chitosan and its biomedical applications", *33rd Annual Conference of the European Society for Biomaterials (ESB-2023)*, 4th – 8th September 2023, Davos, Switzerland.
3. **Amit Kumar**, V.P. Anju, Nitish R. Mahapatra and Pijush Ghosh, "Impact of Tuning Solvent Characteristics on Self-folding of Biopolymer Thin Film", *Active Polymeric Materials and Microsystems (APMM-2019)*, 16th – 19th September 2019, Dresden, Germany.
4. **Amit Kumar**, Smruti Parimita, Nitish R. Mahapatra and Pijush Ghosh, "pH-Responsive Soft Actuator for Biomedical Applications", *3D Graphy Engineering and Medical 2023*, (3DGRAPHY- 2023), 9th - 10th December 2023, IIT Bombay, Maharashtra, India.
5. **Amit Kumar**, Nitish R. Mahapatra and Pijush Ghosh, "pH-responsive bilayer films with engineered non-morphing point of actuation for controlled model drug release", *14th International Conference on Advancements in Polymeric Materials (APM-2023)*, 17th – 19th March 2023, CIPET (SARP)-APDDRL, India.

WORKSHOP ATTENDED

1. Tumor Microenvironment and Resistance to Drugs: The Role of Biomaterials, In Silico and In Vitro Modeling, 19th September 2025, University of Pavia, Italy.
2. Additive Manufacturing of Bio-Implants - Academic & Industry Perspectives, 10th - 11th March 2023, Indian Institute of Technology Madras, India.

RESEARCH AWARDS AND GRANTS

- **MARIE CURIE POSTDOCTORAL FELLOWSHIP- SEAL OF EXCELLENCE** 2026
— My proposal "4DEX-implant" secured 90.80% and was recognized as a high-quality project by the European Commission.
- **RESEARCH EXCELLENCE AWARD- IIT MADRAS** 2024
— Best Research Publication in the Department of Biotechnology, awarded by IIT Madras during Dec 2023- Dec 2024.
- **INSTITUTE RESEARCH AWARD- IIT MADRAS** 2024
— Best Doctoral Research in the Department of Biotechnology, awarded by IIT Madras.
- **FIRST PRIZE- SCI-TECH EXPOSIUM, IIT MADRAS** 2024
— First prize in Oral presentation at Applied Mechanics and Biomedical Engineering (AMBE) Research Palooza.
- **GRADUATE APTITUDE TEST IN ENGINEERING (GATE)- INDIA** 2017
— National level scholarship exam to pursue M.S. (by research) from IIT Madras through HTRA scheme.

POSITION OF RESPONSIBILITY

- **MENTORING- PHD SCHOLAR, IIT ITALY** DEC'25 - PRESENT
— Provided PhD mentorship in designing hydrogel sacrificial mold-based 3D elastomer origami actuators and demonstrating reversible hydrogel swelling-driven actuation for programmable soft robotic systems.
- **WORKSHOP ORGANISER- IEEE RAS ROBOSOFT 2026, KANAZAWA, JAPAN** APRIL'26
— Organized a full-day workshop titled "[Functional and Untethered Soft Machines Powered by the Environment](#)" with the theme "Can soft machines achieve true autonomy by dynamically adapting to environmental changes or by harvesting and utilizing ambient energy sources?". The workshop included: invited talks, a concept competition, a demonstration competition, and a poster competition.
- **GUEST EDITOR, SPECIAL ISSUE, BIOINSPIRATION & BIOMIMETICS, IOP PUBLISHING** PRESENT
— The Special Issue on the topic "[Soft Robots and Actuators Powered by the Environment](#)", seeks to consolidate and advance research at the intersection of soft robotics, bioinspiration, environmental energy harvesting, and sustainable design.
- **TEACHING ASSISTANT- GENETIC ENGINEERING LAB, IIT MADRAS** AUG'19 - DEC'19
— Conducted competent cells, transformation, plasmid isolation, gel electrophoresis, restriction digestion, and PCR amplification experiments.
- **TEACHING ASSISTANT- MICROBIOLOGY LAB, IIT MADRAS** AUG'18 - DEC'18
— Successfully conducted experiments for media preparation, pure culture isolation, Gram staining, and selective media preparation.

EXTRACURRICULAR ACTIVITIES

- Secured 3rd place in both Football and Pole Vault (athletics) at Inter IIT Sports Meet, IIT Gandhinagar in December, 2023.
- Captained Institute Athletics team to secure Runner-Up position at Inter IIT Sports Meet, IIT Delhi in December, 2022.
- Held Inter IIT Meet Record for 3 editions (2018-2022) in Pole Vault with a height of 3.82m and won the Gold Medals for three consecutive years in Inter IIT Sports Meet among 21 IITs across India.

ACADEMIC REFEREES

1. [Prof. Barbara Mazzolai](#), Director, Bioinspired Soft Robotics Lab, Italiano di Tecnologia (IIT), Genoa, Italy
2. [Prof. Nitish R. Mahapatra](#), Professor, Cardiovascular Genetics Lab, Department of Biotechnology, Indian Institute of Technology Madras, India
3. [Prof. Pijush Ghosh](#), Professor, Nanomechanics Lab, Department of Applied Mechanics & Biomedical Engineering, Indian Institute of Technology Madras, India
4. [Dr. Nidhi Gupta](#), Senior Consultant, Dr. Shroff's Charity Eye Hospital, India